Sensorless Control of the BLDC Motors From Near-Zero to High Speeds

Tae-Hyung Kim, *Member, IEEE,* and Mehrdad Ehsani, *Fellow, IEEE*

***Abstract*—This paper presents the theory and implementation of a novel sensorless control technique for the brushless dc (BLDC) motor. The proposed new sensorless drive method solves the problem of the sensorless BLDC motor drives at very low speeds. It provides a highly accurate and robust sensorless operation from near zero to high speeds. For this purpose, an approach, a new flux linkage function is defined, that is speed-independent. The validity of the proposed method is verified through both simulation and experimental results and discussion.**

***Index Terms*—Brushless dc (BLDC) motor, control, sensorless.**

I. Introduction

THE permanent magnet brushless dc (BLDC) motor is increasingly being used in computer, aerospace, military, automotive, industrial and household products because of its high torque, compactness, and high efficiency. The BLDC motor is inherently electronically controlled and requires rotor position information for proper commutations of current. However, the problems of the cost and reliability of rotor position sensors have motivated research in the area of position sensorless BLDC motor drives. Solving this problem effectively will open the way for full penetration of this motor drive into all low cost, high reliability, and large volume applications.

In the last two decades, many sensorless drive solutions have been offered to eliminate the costly and fragile position sensor for BLDC motors with trapezoidal back-EMFs [1]–[10]. The back-EMF voltage sensing [1], [2], back-EMF integration [3], [4], flux estimation [5], [6], detection of the freewheeling diodes conduction [7], and motor modification technique [11] are the main categories of past sensorless solutions for BLDC motors. However, none work well at all speeds without accuracy, reliability, and complexity problems, especially at low speed range. Typically, practical minimum speed of the conventional sensorless drive is around 10% of the rated speed. Also, the position error from a phase shifter in transient state deteriorates the performance of sensorless drive [1], [2], [7]. These drawbacks of sensorless BLDC motor drives have been an obstacle to the use of this motor in various industrial applications.

This paper presents a novel sensorless position detection technique with a new physical concept based on a speed-independent position function for the BLDC motors. A physically insightful speed-independent function of flux linkage, along with the combination of two differential equations governing the stator phase windings, has been used for this purpose. With the speed-independent position function, the commutation instants can be estimated from near zero (1.5% of the rated speed) to high speeds. Since the shape of the position function is identical at all speeds, it provides a precise commutation pulse at steady state as well as transient state. The proposed method does not rely on the measured back-EMF; hence the need for external hardware circuitry for sensing terminal voltages has been removed. Moreover, the proposed sensorless algorithm is very suitable for implementation on low cost, fixed point digital signal processors (DSPs). A simulation model for the sensorless BLDC motor drive has been developed to validate the proposed methods, and experimental results are shown to prove the methods.

Manuscript received March 14, 2003; revised November 21, 2003. This work was presented at the Applied Power Electronics Conference and Exposition, 2003. Recommended by Associate Editor J. Ojo.

The authors are with the Department of Electrical Engineering, Texas A&M University, College Station, TX 77843-3128 USA (e-mail: thkim@ee.tamu.edu).

Digital Object Identifier 10.1109/TPEL.2004.836625

II. Review of Sensorless Control Methods for BLDC Motors

Sensorless BLDC Motor control has been a research topic for the last two decades. Many papers and patents have been published on this topic [9]. The main methods published in literature on this topic especially for the trapezoidal back-EMF type of BLDC motors can be classified as follows.

A. Back-EMF Sensing Techniques [1], [2]

The BLDC motor has a trapezoidal shape of the induced back-EMF in the stator winding. Monitoring the phase back-EMF measured from terminal voltages in the silent phase, the zero crossing of the back-EMF can be detected. Since back-EMF is zero at standstill and proportional to speed, the measured terminal voltage that has large signal-to-noise ratio cannot detect zero crossing at low speeds. Also, the estimated commutation points that are shifted by 30° from zero crossing of back-EMFs have position error in transient state. With terminal voltage-sensing method [1], an operating speed range is typically around 1000–6000 RPM. The third harmonic back-EMF sensing method [2] provides wider speed range than the terminal voltage-sensing method.

B. Back-EMF Integration Technique [3], [4]

In this method, position information is extracted by integrating the back-EMF of the silent phase. Integration starts when the open phase's back-EMF crosses zero. A threshold is set to stop the integration that corresponds to a commutation instant. This method also has a problem at low speeds because of the error accumulation problem.

0885-8993/04$20.00 © 2004 IEEE

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

C. Flux Linkage-Based Technique [5], [6]

In this method, the flux linkage is calculated using measured voltages and currents. The fundamental idea is to take the voltage equation of the machine and by integrating the applied voltage and current, flux can be estimated. From the initial position, machine parameters, and the flux linkages' relationship to rotor position, the rotor position can be estimated. This method also has significant estimation error in low speed. Improper error of parameters and sampled current is reason for accumulation error at low speeds in which the voltage equation is integrated in a relatively large period of time.

D. Freewheeling Diode Conduction [7]

This method uses current flowing through a freewheeling diode in silent phase. For a short period after reaching zero crossing of the back-EMF in silent phase, a tiny current is flowing through freewheeling diode; during the active phase switches are turned off under alternate chopper control. This silent phase current starts to flow in the middle of the commutation interval, which corresponds to the point where back-EMF of the open phase crosses zero. This method also has position error of commutation points in transient state. The most serious drawback of this method is the requirement of six additional isolated power supplies for the comparator circuitry to detect current flowing in each freewheeling diode.

III. Proposed Sensorless Control Method

Most popular and practical sensorless drive methods for BLDC motors rely on speed-dependent back-EMF. Since the back-EMF is zero or undetectably small at standstill and low speeds, it is not possible to use the back-EMF sensing method in the low speed range. Also, the estimated commutation points that are shifted by 30° from zero crossing of back-EMFs have position error in transient state. The flux estimation method [5], [6] also has significant estimation error at low speed, in which the voltage equation is integrated in a relatively large period of time. In this paper, to overcome the above drawbacks, a novel method, based on a new speed-independent function, is proposed, one that is based on a new physical insight.

A. Principle of the Proposed Method

Each active phase in an ac motor can be described by a first order differential equation. The general voltage equation of one of the active phases is given by

$$v_x = Ri_x + \sum_{k=1}^{n} \frac{d\psi_{kx}(\theta, i_x)}{dt} \tag{1}$$

where v_x is the active phase voltage, R is the phase resistance, i_x is the phase current, θ is the rotor position, $\psi_{kx}(\theta, i_x)$ is the total flux linkage of the active phase, and "n" is the number of phases. The flux linkage in the active phase includes both self and mutual flux linkages. For three-phase BLDC motors, the total flux linkage of the phase A is

$$\psi_A = L_{\rm aa}(\theta, i_a)\cdot i_a + L_{\rm ab}(\theta, i_b) \cdot i_b + L_{\rm ac}(\theta, i_c)\cdot i_c + \lambda_{\rm ar}(\theta). \tag{2}$$

Where, the first term, $L_{\rm aa}(\theta, i_a)\cdot i_a$, represents the self flux linkage of phase A; the second and third term stand for the mutual flux linkage with phases B and C; and the fourth term, $\lambda_{\rm ar}(\theta)$, stands for the flux linkage due to the permanent magnet that is attached on the rotor. The first three terms of (2) are function of current and position, and the $\lambda_{\rm ar}(\theta)$ is the function of position. Therefore, the flux profile has a close relationship with the dynamic performances. Since the BLDC motor uses permanent magnets, the permeability of the attached permanent magnet is considered as that of air, and hence, typically, the motor has small inductance variation. Therefore, if the motor is operated within the rated current, the saturation effect of inductance due to current level is usually neglected. For the surface-mounted permanent magnet (SMPM) type of BLDC motors, the permanent magnets are roundly attached on the round surface of the rotor, and hence, the inductance variation by rotor position is negligibly small. Based on the characteristics of the SMPM type of BLDC motors, the flux profile can be simplified as (3) along with the following assumptions:

1) the motor is operated within the rated condition and hence the saturation effect due to current level is neglected;
2) the leakage inductance is negligibly small and hence neglected;
3) iron losses are negligible;

$$\psi_A = L_{\rm aa}\cdot i_a + L_{\rm ab}\cdot i_b + L_{\rm ac}\cdot i_c + \lambda_{\rm ar}(\theta). \tag{3}$$

Substituting (3) into (1) gives

$$v_a = R_a i_a + \frac{d}{dt}(L_{\rm aa}i_a + L_{\rm ab}i_b + L_{\rm ac}i_c) + \frac{d\lambda_{\rm ar}(\theta)}{dt}. \tag{4}$$

In balanced three-phase BLDC motors

$$\begin{aligned} &L_{\rm aa} = L_{\rm bb} = L_{\rm cc} = L_s, \\ &L_{\rm ba} = L_{\rm ab} = L_{\rm ca} = L_{\rm ac} = L_{\rm bc} = L_{\rm cb} = L_m. \end{aligned} \tag{5}$$

Where, L_S, and L_m represent the self-inductance and mutual-inductance, respectively. Substituting (5) into (4) gives

$$v_a = R_a i_a + \frac{d}{dt}(L_s i_a + L_m i_b + L_m i_c) + \frac{d\lambda_{\rm ar}(\theta)}{dt}. \tag{6}$$

For a balanced wye-connected BLDC motors, the three-phase currents always meet the following equation:

$$i_a + i_b + i_c = 0. \tag{7}$$

Using (7), the (6) is simplified as

$$\begin{aligned} v_a &= R_a i_a + (L_s - L_m)\frac{di_a}{dt} + \frac{d\lambda_{\rm ar}(\theta)}{dt} \\ &= R_a i_a + L\frac{di_a}{dt} + \frac{d\lambda_{\rm ar}(\theta)}{dt} \quad (L = L_s - L_m) \end{aligned} \tag{8}$$

where L represents the phase inductance under balanced conditions. The last term in the voltage equations is so called back-EMF, and the term is redefined as

$$\begin{aligned} v_a &= Ri_a + L\frac{di_a}{dt} + \frac{d\lambda_{\rm ar}(\theta)}{dt} = Ri_a + L\frac{di_a}{dt} + \frac{d(k_e f_{\rm ar}(\theta))}{dt} \\ &= Ri_a + L\frac{di_a}{dt} + k_e\cdot\frac{d\theta}{dt}\cdot\frac{d(f_{\rm ar}(\theta))}{d\theta} \end{aligned} \tag{9}$$

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

where k_e stands for the back-EMF constant. It is seen that $\lambda_{\rm ar}(\theta)$ in (8) is expressed as a constant value times a flux linkage function that is changed only by the rotor position as shown in (9). The $f_{\rm ar}(\theta)$ is a flux linkage form function that is a function of rotor position. Since some manufacturers do not provide a motor neutral point, the line-to-line voltage equations are used as follows:

$$V_{\rm ab} = R(i_a - i_b) + L\frac{d(i_a - i_b)}{dt} + k_e \cdot \frac{d\theta}{dt} \cdot \frac{d\{f_{\rm ar}(\theta) - f_{\rm br}(\theta)\}}{d\theta} \quad (10)$$

$$V_{\rm ab} = R(i_a - i_b) + L\frac{d(i_a - i_b)}{dt} + k_e \cdot \frac{d\theta}{dt} \cdot \frac{df_{\rm abr}(\theta)}{d\theta} \quad (11)$$

$$V_{\rm ab} = R(i_a - i_b) + L\frac{d(i_a - i_b)}{dt} + k_e \cdot \omega \cdot \frac{df_{\rm abr}(\theta)}{d\theta} \quad (12)$$

where ω stands for the instantaneous speed. The $f_{\rm abr}(\theta)$ is a line-to-line flux linkage form function that is a function of the rotor position. Now we define a new function, $H(\theta)_{\rm ab}$, as

$$H(\theta)_{\rm ab} = \frac{df_{\rm abr}(\theta)}{d\theta}. \quad (13)$$

Then, $H(\theta)_{\rm ab}$ can be derived as

$$H(\theta)_{\rm ab} = \frac{1}{\omega \cdot k_e}\left[(V_a - V_b) - R(i_a - i_b) - L\left(\frac{di_a}{dt} - \frac{di_b}{dt}\right)\right]. \quad (14)$$

Since the $H(\theta)_{\rm ab}$ function itself has a one to one relationship with rotor position, it is possible to use this function for position estimation. But as shown in (14), to know the $H(\theta)_{\rm ab}$ function, the instantaneous speed term, that is unknown for dynamic operations, is required to calculate the function.

B. Derivation of the Speed-Independent Position Function

To eliminate the instantaneous speed term, ω, that causes trouble in using the $H(\theta)_{\rm ab}$ function for position estimation, one line-to-line $H(\theta)$ function is divided by another line-to-line $H(\theta)$ function, and the divided new speed independent function is named $G(\theta)$. For example

$$\begin{aligned}\frac{H(\theta)_{\rm bc}}{H(\theta)_{\rm ab}} &= \frac{\frac{1}{\omega \cdot k_e}\left[(V_b - v_c) - R(i_b - i_c) - L\left(\frac{di_b}{dt} - \frac{di_c}{dt}\right)\right]}{\frac{1}{\omega \cdot k_e}\left[(V_a - V_b) - R(i_a - i_b) - L\left(\frac{di_a}{dt} - \frac{di_b}{dt}\right)\right]} \\ &= \frac{\left[(V_b - V_c) - R(i_b - i_c) - L\left(\frac{di_b}{dt} - \frac{di_c}{dt}\right)\right]}{\left[(V_a - V_b) - R(i_a - i_b) - L\left(\frac{di_a}{dt} - \frac{di_b}{dt}\right)\right]} \\ &= G(\theta)_{\rm bc/ab}. \end{aligned} \quad (15)$$

The digitized expression of the function is (16), shown at the bottom of the page.

Where, $S_a(k)$, $S_b(k)$, and $S_c(k)$ are switching status of phase A, B, and C and known values by the controller. Fig. 1 shows the

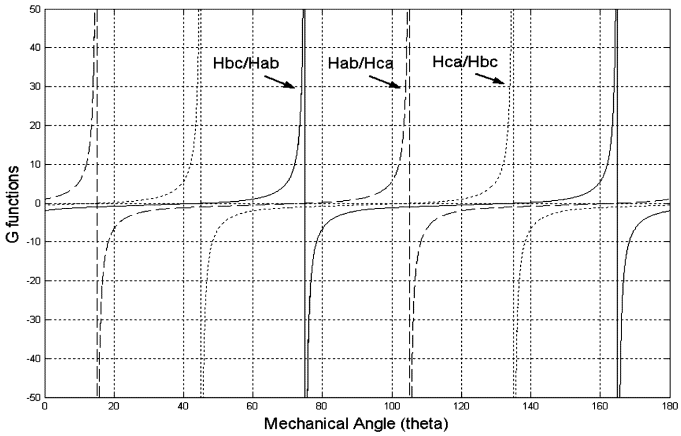


Fig. 1. G function waveforms.

$G(\theta)$ functions ($G(\theta)_{bc/ab}, G(\theta)_{ab/ca}$, and $G(\theta)_{ca/bc}$) waveform based on (15). The standard commutation instant is when the $G(\theta)$ functions are changed from positive infinity to negative infinity as in Fig. 1.

The dc link bus voltage, $V_{\rm bus}$ and currents are sampled with 40-kHz sampling frequency. To avoid the deterioration of the G function signal due to the noise from measured values, the sparse calculation method is applied. The method is especially useful at very low speeds such as 0.25 Hz (15 RPM). The number of sampled data in the 60° commutation period reaches to more than 10 000 at around 15 RPM. If the calculation frequency for G function is set to have 0.5° resolution, one calculation cycle can include more than 100 sampled data. The collected data in one cycle are arrived at one sample which is time average of the data to decrease noise in the calculation of $V_{\rm bus}$, phase currents, and di/dt terms. To polish the critical values in the G function especially at low speeds, the calculation frequency is modified proportionate to speed to collect more data at low speeds. The calculation frequency is changed to maintain 0.5° resolution while the speed is changing. This delay due to the sparse calculation can cause the deterioration of the acceleration or deceleration performance. The speed-independent function method along with the sparse calculation provides 15 Hz/s acceleration or deceleration performance.

C. Commutation Strategy

After careful investigation of the shape of $G(\theta)$ function, to provide the best indirect position-function that is speed-independent with high sensitivity at each commutation point, the ratio of two line-to-line $H(\theta)$ functions is sequentially utilized. From the sequential combination of two line-to-line $H(\theta)$ functions at each commutation interval, the ideal $G(\theta)$ function that

$$G(\theta)_{\rm bc/ab(k)} = \frac{V_{\rm bus} \cdot (S_b(k) - S_c(k)) - R \cdot (i_b(k) - i_c(k)) - L \cdot \left(\frac{(i_b(k) - i_b(k-1)) - (i_c(k) - i_c(k-1))}{t_k - t_{k-1}}\right)}{V_{\rm bus} \cdot (S_a(k) - S_b(k)) - R \cdot (i_a(k) - i_b(k)) - L \cdot \left(\frac{(i_a(k) - i_a(k-1)) - (i_b(k) - i_b(k-1))}{t_k - t_{k-1}}\right)} \quad (16)$$

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

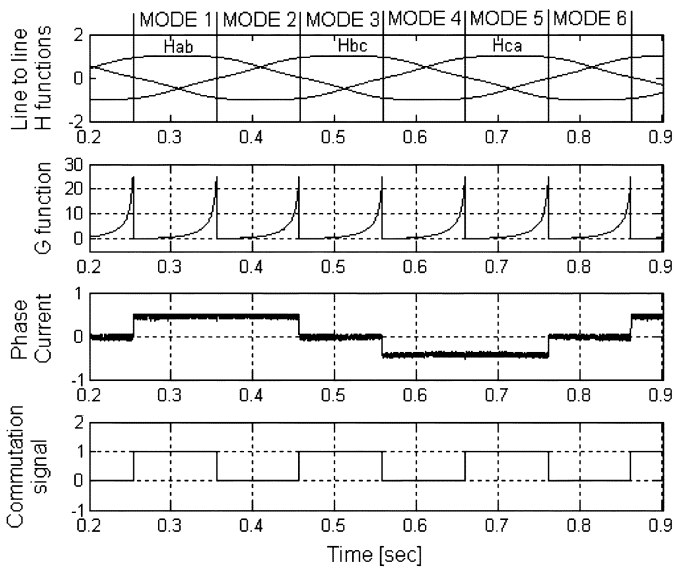


Fig. 2. Sensorless operation using the proposed method (from top to bottom: line-to-line H functions; G function; phase current; generated commutation signal).

TABLE I
G FUNCTION AT EACH MODE

Mode 1 and 4	$G(\theta)_1 = \frac{H(\theta)_{ca}}{H(\theta)_{bc}} = \frac{V_{ca} + Ri_{ca} + L\frac{di_{ca}}{dt}}{V_{bc} - Ri_{bc} - L\frac{di_{bc}}{dt}}$
Mode 2 and 5	$G(\theta)_2 = \frac{H(\theta)_{bc}}{H(\theta)_{ab}} = \frac{V_{bc} + Ri_{bc} + L\frac{di_{bc}}{dt}}{V_{ab} - Ri_{ab} - L\frac{di_{ab}}{dt}}$
Mode 3 and 6	$G(\theta)_3 = \frac{H(\theta)_{ab}}{H(\theta)_{ca}} = \frac{V_{ab} + Ri_{ab} + L\frac{di_{ab}}{dt}}{V_{ca} - Ri_{ca} - L\frac{di_{ca}}{dt}}$

is independent of speed and contains continuous position information is derived. It means that only the positive part of the $G(\theta)$ function waveform in Fig. 1 is utilized. It is noted that the standard commutation point is the peak point of ideal $G(\theta)$ function that is the most sensitive part of the function as shown in Fig. 2. Therefore, the position estimation based on the $G(\theta)$ function can provide the best accuracy at commutation points of the current which are the most important instants for the sensorless operation.

Table I shows the position estimation equations at each mode in Fig. 2. Two line-to-line $H(\theta)$ functions are used in each mode as Table I. As shown in Fig. 2, at mode 1, $G(\theta)_1$ is used as a position estimation equation; after a 60° electrical angle, at mode 2, $G(\theta)_2$ is utilized. The time duration of each mode in Fig. 2 corresponds to 60° by electrical angles.

When the $G(\theta)$ function reaches a predefined threshold value, the motor is commutated. The threshold value is defined based on the current rising time and desired advanced angle. From the sequential combination of the $H(\theta)$ functions based on Table I, the $G(\theta)$ function can be made as shown in Fig. 2. It is noted

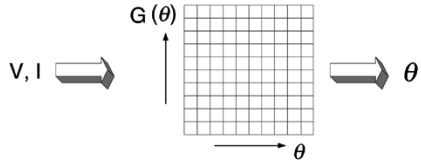


Fig. 3. Estimation of the position using the characterized G function.

that the commutation signal can be generated at the peak point that is the most sensitive part of the $G(\theta)$ function. From the available stator current and dc link voltage, the $G(\theta)$ function is computed based on the equations of Table I in real time.

Since the waveform of the $G(\theta)$ function is identical at the entire speed range, it can be characterized at steady state in a look-up table, and used as a position reference for sensorless operation at all speeds. If the control purpose is just to follow a square wave reference current for constant torque, the commutation points that are detected at the peak point of the $G(\theta)$ function are sufficient as position information, but for special purpose control strategies, for example, an advanced angle control for the field weakening operation, it is required to know more position information between commutation points.

Fig. 3 shows the procedure to estimate position between commutation points. From the measured current and line-to-line voltages that are calculated by multiplication of the dc bus voltage and switching status, $G(\theta)$ function is computed at each mode. Then, position is estimated by the look-up table of the characterized $G(\theta)$ function.

D. Current Control

To control currents, a simple PI or hysteresis controller can be used for the proposed sensorless method. Also, to calculate line-to-line voltages in equations of Table I at each mode, the three phase activated current control is utilized. It means that a silent phase current is controlled as zero.

E. Starting Procedure

In typical sensorless operation of the BLDC motor, the forced alignment of the rotor is a way of setting an initial position [1]–[4]. Two phases are excited to cause the rotor to shift and lock into position. If the rotor is not in the desired position, the forcing torque from the excited phases causes it to rotate and stop at the desired position. After energizing two of the three motor phases for enough time to ensure the rotor will lock into position, the next commutation signal advancing the switching pattern by 60° is given, then immediately the proposed sensorless algorithm using the $G(\theta)$ function is applied to detect the next commutation instant. The 60° rotor movement is enough to detect the commutation instant using the position estimation equations in Table I. After the first detection of the commutation point, both torque and speed control is possible using the estimated speed from time duration of each commutation point.

IV. SIMULATION RESULTS

The proposed sensorless control algorithm has been verified through numerical simulation. Figs. 4 and 5 illustrate the performance of the proposed sensorless method at 50 and 1000 RPM,

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

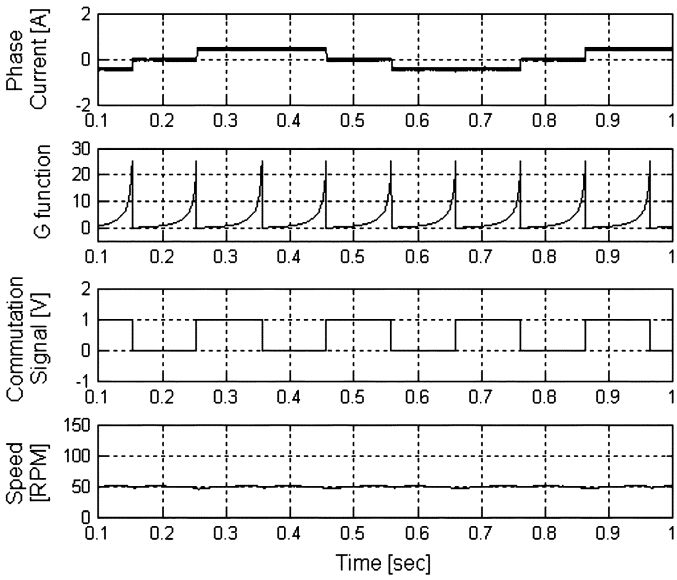


Fig. 4. Simulation results of the proposed sensorless method at 50 RPM (from top to bottom: phase current; G function; virtual commutation signal; speed).

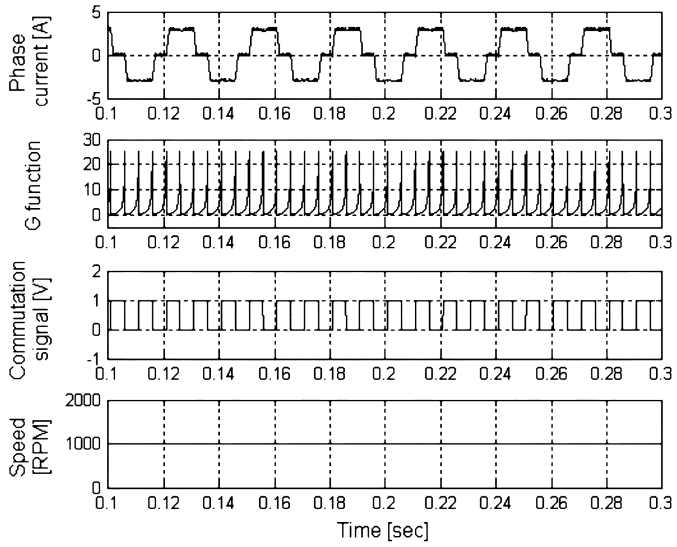


Fig. 5. Simulation results of the proposed sensorless method at 1000 RPM (from top to bottom: phase current; G function; virtual commutation signal; speed).

respectively. The phase current can be controlled by any control strategy such as PI or hysteresis controller. In this simulation, each phase current is controlled with the hysteresis control strategy.

The back-EMF, $(d\lambda_{\rm ar}(\theta)/dt)$, is modeled for simulation using Fourier series based on the measurement of BLDC motor that is used in experiment as

$$\frac{d\lambda_{\rm ar}(\theta)}{dt} = 2 \cdot B_g \cdot r \cdot l \cdot \omega_r \cdot N_a \cdot \sum_{n=1}^{\infty} A_n \cdot \cos[n(\theta)]. \quad (17)$$

For fast simulation, only the first three terms of the Fourier series are utilized to construct the back-EMF profile as

$$\frac{d\lambda_{\rm ar}(\theta)}{dt} = 2 \cdot B_g \cdot r \cdot l \cdot \omega_r \cdot N_a \cdot [A_1 \cdot \cos(\theta) + A_3 \cdot \cos(3\theta) + A_5 \cdot \cos(5\theta)]. \quad (18)$$

where $B_g, r, l, \omega_r, N_a, A_n$, and θ are the air-gap flux density, rotor radius, length, speed, number of turns for phase A, n_{th} harmonic coefficient, and rotor position, respectively.

The current commutation is performed when the $G(\theta)$ function reaches a predefined threshold value. The threshold value can be modified for proper advanced angle control. Based on the $G(\theta)$ function, the virtual commutation signal is generated as in Figs. 4 and 5.

Fig. 6 shows the starting performance and speed response of the proposed sensorless control method. Fig. 6(a) and (b) show the excited current and speed waveform for the forced alignment procedure and transient state. The currents are initially excited from phase A to B and controlled with the hysteresis control strategy. From zero to 3.5 s, the motor is locked into a position and the next commutation signal is given to run the motor and to detect the rotor position. During the first 60° periods, the $G(\theta)$ function is computed to estimate the rotor position and detect the first commutation point. Fig. 6(c) shows the $G(\theta)$ function waveform for the starting of the motor. Even in transient state, precise commutation points can be obtained by the $G(\theta)$ function. The comparison between the virtual Hall sensor that is generated by the $G(\theta)$ function and the real Hall sensor in transient state is presented in Fig. 6(d).

V. DSP-Based Implementation and Experimental Results

Fig. 7 shows an overall system block diagram of the proposed new sensorless drive. In this experiment, the PI controller is used for each phase current control. The phase terminal voltages are calculated based on the dc link voltage, switching status of the inverter, and voltage drops of the inverter switches. When the dc link voltage has a large ripple due to the size of a dc link capacitance, a voltage sensor is required to sense the dc link bus voltage.

The experimental setup consists of a 1-HP, 5.5-A, 1000 RPM, four-pole BLDC motor drive loaded with a permanent magnet dc generator as shown in Fig. 8. The entire drive system is controlled by a low-cost, fixed-point digital signal processor (DSP), TMS320F243.

Table II shows the four-pole BLDC motor specification to examine the performance of the developed sensorless drive technique.

Figs. 9 and 10 show the experimental waveforms of the $G(\theta)$ function, virtual Hall sensor signal made by the $G(\theta)$ function, phase current, and speed, respectively, at 50 and 20 RPM. Also, Fig. 11 shows the phase current, $G(\theta)$ function, and speed waveform at 900 RPM. The $G(\theta)$ function is computed in the DSP and sent to a D/A (digital/analog) converter to observe it through an oscilloscope.

To observe the accuracy of estimated commutation points, the virtual Hall sensor signal that is generated by the $G(\theta)$ function is compared with the real Hall sensor signal as in Fig. 12.

Each edge indicates the commutation instant. It is noted that the position estimation error of the virtual Hall sensor is less than 5° by electrical angle even at this low speed.

Fig. 13 shows the starting performance of the proposed sensorless method. The machine is initially aligned by exciting two

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

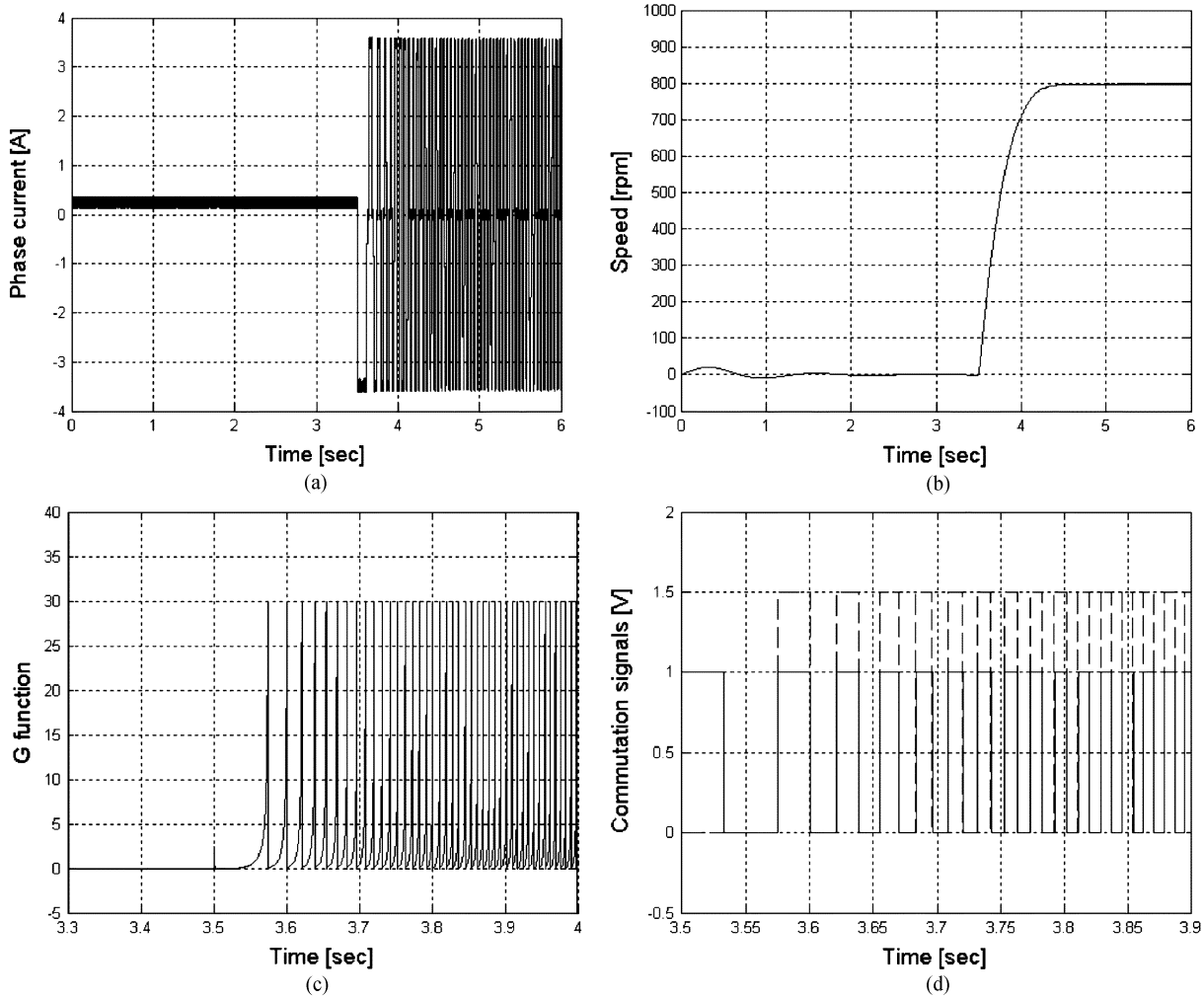


Fig. 6. Speed response and performance for the starting and transient state (current control mode): (a) phase current, (b) speed response, (c) G function, and (d) virtual and real Hall sensor signals.

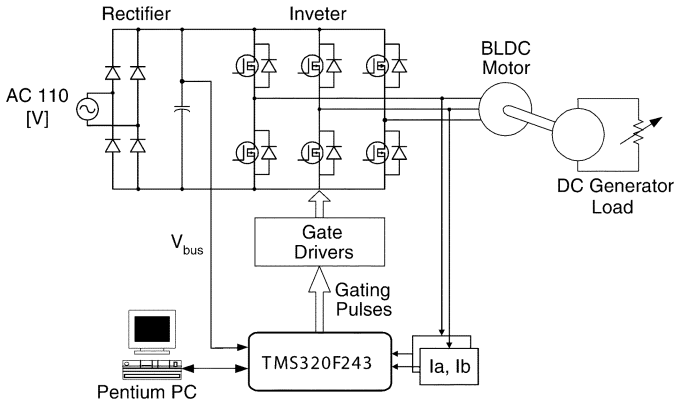
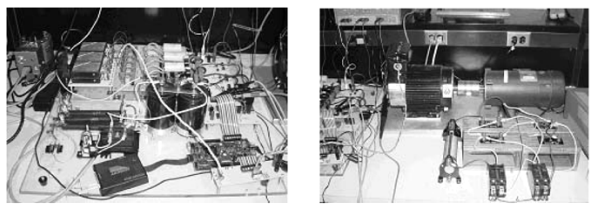


Fig. 7. Block diagram of the proposed sensorless drive.

TABLE II
MOTOR SPECIFICATION (LL: LINE-TO-LINE)

Power	1 HP
Rated Speed	1000 [RPM]
$R_{_LL}$	8.6 [Ω]
$L_{_LL}$	77.6 [mH]
K_e	1.48 [V/(rad/sec)]
K_t	1.605 [Nm/A]



(a) (b)

Fig. 8. Experimental test-bed: (a) overall system and (b) BLDC motor and dc generator load.

phases from zero to 2 s as in Fig. 13. After the forced alignment, during the first 60° commutation interval, the $G(\theta)$ function is computed and estimate commutation points as Fig. 13. The starting procedure from standstill to 50 RPM as in Fig. 13 is smooth and has no serious distortion of speed.

Fig. 14 shows the performance of the sensorless operation under a sudden load change. As shown in Fig. 7, a dc generator is used as a load and connected by a resistance bank. The motor torque is controlled by current control, and from a step change of the resistance value at a certain time, the load is changed.

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

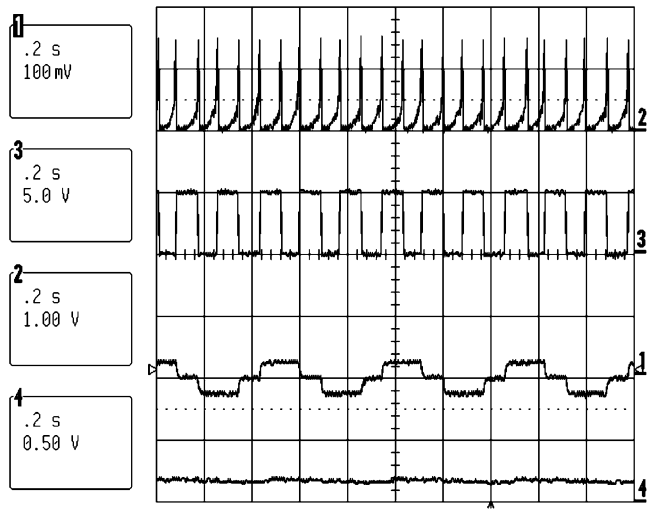


Fig. 9. Sensorless operation at 50 RPM [200 ms/div.] (from top to bottom: G function; virtual Hall sensor signal [5 V/div.]; phase current [2 A/div.]; speed [125 RPM/div.]).

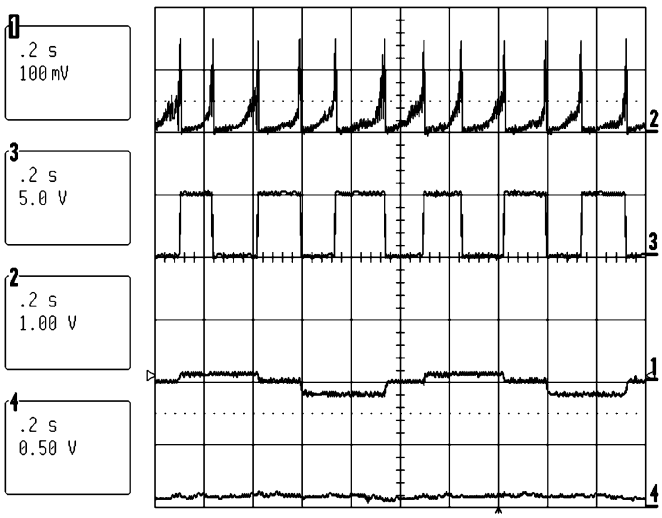


Fig. 10. Sensorless operation at 20 RPM [200 ms/div.] (from top to bottom: G function; virtual Hall sensor signal [5 V/div.]; phase current [2 A/div.]; speed [125 RPM/div.]).

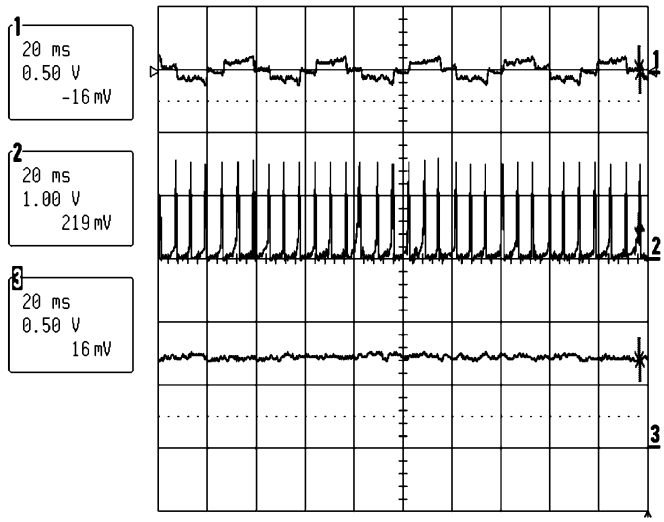


Fig. 11. Sensorless operation at 900 RPM [20 ms/div.] (from top to bottom: phase current [10 A/div.]; G function; speed [600 RPM/div.]).

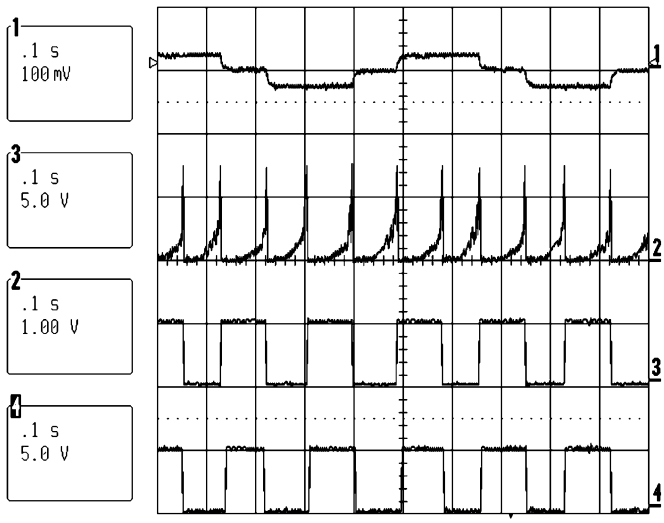


Fig. 12. Comparison of the virtual Hall sensor signal and real Hall sensor signals at 50 RPM [100 ms/div.] (from top to bottom: phase current [2 A/div.]; G function; virtual Hall sensor signal [5 V/div.]; real Hall sensor signals [5 V/div.]).

In this case, the motor speed is changed from 50 to 180 RPM. Under the load change, in transient period, reliable sensorless operation is guaranteed based on the speed-independent $G(\theta)$ function.

Theoretically, the speed independent function is independent with the load variation if the operational speed is bigger than the stated minimum speed. But, in practical operation, the sensorless drive has the acceleration/deceleration performance such as 15 Hz/s (mechanical frequency) due to the limited resolution caused by calculation frequency. The sensorless drive can handle load changes that make less speed variation than 15 Hz/s.

VI. Estimation Error of the Proposed Method

The results of the position estimation error are the ripple of total torque and decrease of the available torque per commutation period, hence, the decrease of torque/ampere ratio. This, in turn, causes a decrease of the motor efficiency. The instantaneous torque and power equations of BLDC motors are

$$T = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega} \ [\mathrm{N}\cdot\mathrm{m}]$$
$$P = e_a i_a + e_b i_b + e_c i_c [\mathrm{W}]. \tag{19}$$

As an example, Fig. 15 shows the ideal shape of back-emf, current, and total torque under commutation point error (θ_e) in indirect position sensing of BLDC motors. Based on Fig. 15, the maximum average torque and power per electrical cycle without any position errors are calculated as

$$T_{\mathrm{av,max}} = \frac{2EI}{\omega}, \quad P_{\mathrm{av,max}} = 2EI \tag{20}$$
$$E = k_e \cdot \omega \tag{21}$$

where k_e represents the back-EMF constant.

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

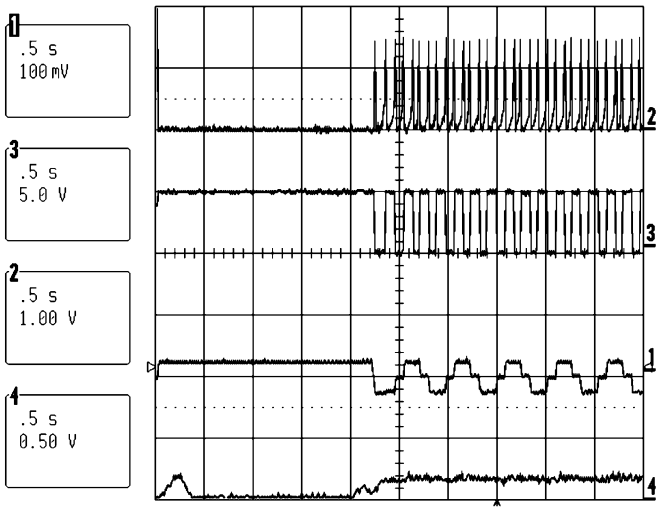


Fig. 13. Sensorless starting [0.5 s/div.] (from top to bottom: G function; virtual Hall sensor signal [5 V/div.]; phase current [2 A/div.]; speed [125 RPM/div.]).

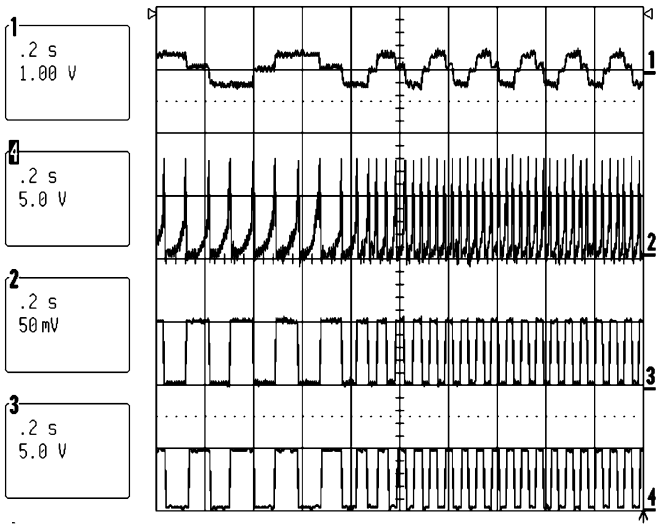


Fig. 14. Performance under sudden load change [200 ms/div.] (from top to bottom: phase current [2 A/div.]; G function; virtual Hall sensor signal [5 V/div.]; real Hall sensor signal [5 V/div.]).

If there is a commutation angle error, θ_e, the available torque per commutation period and motor efficiency under the commutation angle error can be calculated as Table III.

To calculate the estimated position error, the numerator and denominator of (16) can be represented with possible error sources as follows.

1) Numerator with possible error sources as (22), shown at the bottom of the page.
2) Denominator with possible error sources as (23), shown at the bottom of the next page, where the error sources are

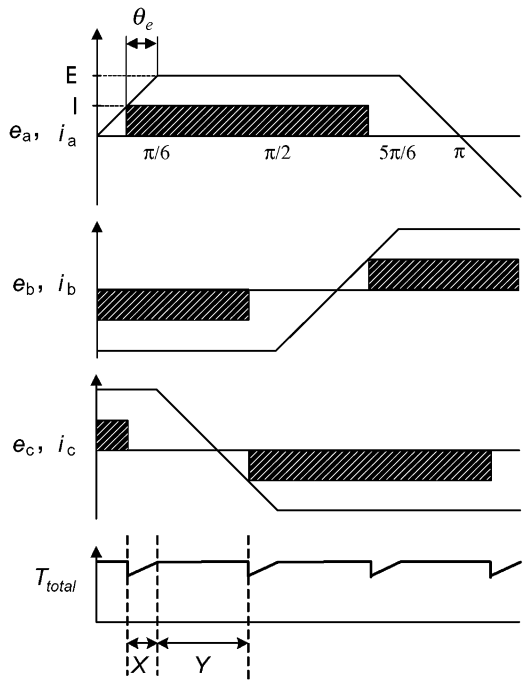


Fig. 15. Effect of rotor position estimation error on the electrical torque: (a) available torque and a load line and (b) three-dimensional (3-D) waveform.

TABLE III
AVAILABLE TORQUE PER COMMUTATION PERIOD

Available Torque Per Commutation Period [N-m]	$0 \le \lvert\theta_e\rvert \le \frac{\pi}{3}$	$2 \cdot k_e \cdot I - \left(\frac{3}{\pi}\right)^2 \cdot k_e \cdot I \cdot \theta_e^2$
	$\frac{\pi}{3} < \lvert\theta_e\rvert \le \frac{2\pi}{3}$	$\left(\frac{3}{\pi}\right) \cdot k_e \cdot I \cdot \left(\pi - 2 \cdot \lvert\theta_e\rvert\right)$

η_1	dc link voltage measurement error $(\text{noise} + \text{sensor error})$;
η_2	Parameter error (phase resistance);
η_5	Parameter error (phase inductance);
$\eta_3, \eta_4, \eta_6, \eta_7, \eta_8,$ and η_9	Current measurement error $(\text{noise} + \text{sensor error})$.

Based on the above error sources, the worst-case numerator and denominator are simulated and compared with the numerator and denominator that do not contain the error sources. In this case, the sign of each error sources is defined to make the worst scenario. With this approach, the estimated position error can be calculated based on the amount of each error sources.

If the worst-case position error is calculated, the available torque per commutation period can be calculated based on the Table III. As an example, Fig. 16(a) shows

$$V_{\text{DC bus}} \cdot (1 \pm \eta_1) \cdot (S_b - S_c) - R \cdot (1 \pm \eta_2) \cdot \left(i_{b(k)} \cdot (1 \pm \eta_3) - i_{c(k)} \cdot (1 \pm \eta_4)\right) - L \cdot (1 \pm \eta_5) \cdot \left(\frac{\left(i_{b(k)} \cdot (1 \pm \eta_3) - i_{b(k-1)} \cdot (1 \pm \eta_6)\right) - \left(i_{c(k)} \cdot (1 \pm \eta_4) - i_{c(k-1)} \cdot (1 \pm \eta_7)\right)}{t_k - t_{k-1}}\right) \quad (22)$$

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

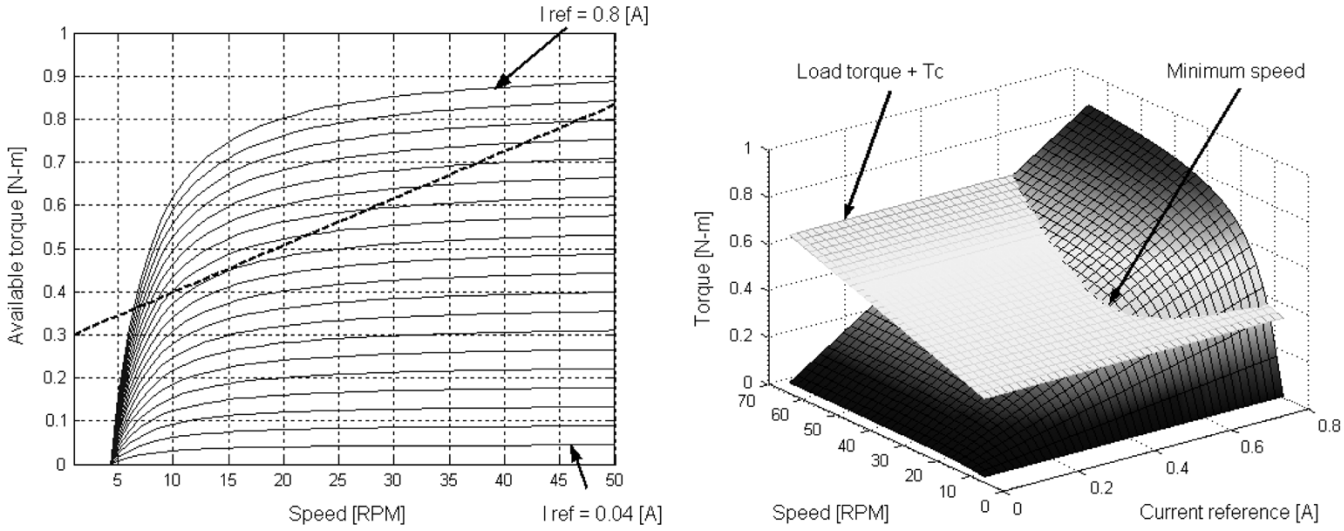


Fig. 16. Available torque per commutation period based on the current reference with a dc generator load.

the available torque and speed characteristic with assumed error sources (current measurement error $= \pm 0.05$ A, dc link voltage measurement error $= \pm 1.0$ V, resistance variation $= \pm 5\%$, inductance variation $= \pm 5\%$) when different current references are given. To define the minimum speed of the sensorless drive method under possible error sources, the characteristics of the electromagnetic torque, load torque, and cogging torque of the BLDC motor are carefully investigated. The cogging torque is the function of rotor position. Since the cogging torque can oppose the electromagnetic torque at certain position, the maximum value of cogging torque is added to the load torque and compared with the electromagnetic available torque to define the speed limitation as shown in Fig. 16(a). A dc generator is used as load in the experimental test-bed. In this case, the load torque has a linear relationship with the motor speed. The dotted straight line presents the dc generator load plus magnitude (maximum value) of the cogging torque.

The intersected points in Fig. 16(a) indicate the operating points of the sensorless drive. In this case, the minimum speed is at around 15 RPM where the dotted line becomes a tangential line of an available torque curve. Fig. 16(b) shows the 3-D waveform of Fig. 16(a). The available electromagnetic torque and load torque surfaces are shown in 3-D space. The intersected line of two surfaces shows the operating speed when the different current command is given. With these assumed error sources, the minimum speed is at around 15 RPM.

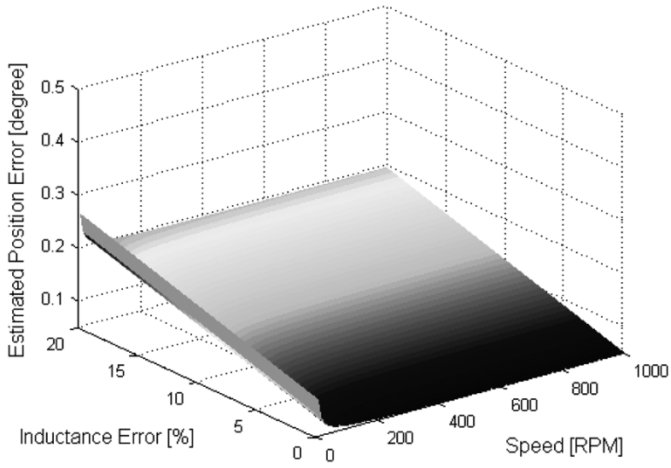


Fig. 17. Impact of inductance error on the position estimation error.

In the proposed method, the constant inductance value is assumed and used for the G function calculation. In actual case, the inductance value is varying by rotor position having maximum and minimum value because of the local saturation. However, for the SMPM type of BLDC motors, the variation is usually small (less than 20% of the dc offset value). Authors used the dc value (average value) of the inductance that is initially measured for the G function calculation. The resultant position estimation error due to the constant inductance value is shown in 3-D space as Fig. 17 based on (19) and (20).

$$V_{\mathrm{DC\,bus}} \cdot (1 \pm \eta_1) \cdot (S_a - S_b) - R \cdot (1 \pm \eta_2) \cdot \left(i_{a(k)} \cdot (1 \pm \eta_8) - i_{b(k)} \cdot (1 \pm \eta_3)\right) - L \cdot (1 \pm \eta_5) \cdot \left(\frac{\left(i_{a(k)} \cdot (1 \pm \eta_8) - i_{a(k-1)} \cdot (1 \pm \eta_9)\right) - \left(i_{b(k)} \cdot (1 \pm \eta_3) - i_{b(k-1)} \cdot (1 \pm \eta_6)\right)}{t_k - t_{k-1}} \right) \quad (23)$$

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.

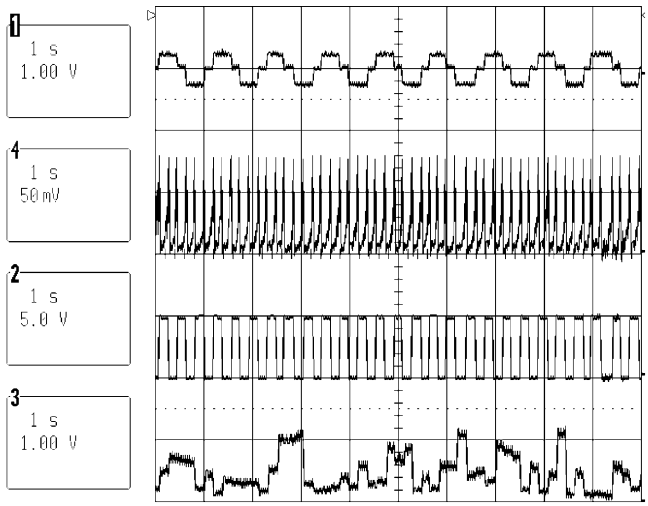


Fig. 18. Commutation angle error at 20 RPM (from top to bottom: phase current [1 A/div.]; G function; virtual Hall sensor signal [5 V/div.]; absolute value of the commutation angle error [2.5° (mechanical)/div.]).

TABLE IV
WORST-CASE COMMUTATION ANGLE ERROR (EXPERIMENTAL RESULT)

Speed [RPM]	20	50	150	300	900
Position error (mechanical angle)	3.5°	2.5°	2.1°	1.9°	1.5°

The position error is less than 0.3° by mechanical angle in the speed range from 20 to 1000 RPM if the applied current is less than rated value. The position error is calculated when the sparse calculation method is utilized. As shown in Fig. 17, the impact of the inductance value error (below 20%) on the proposed position estimation method is negligible.

In Fig. 18, the experimental results about the commutation angle error using the experimental test-bed are shown. The bottom signal in Fig. 18 shows the absolute value of the commutation angle error at 20 RPM. The worst commutation angle errors at different speeds based on the experimental results are shown in Table IV. To find the value of worst-case at each speed, the motor is operated for a long time with the sensorless algorithm, and the worst value of the commutation angle error is stored in a memory of the DSP. The stored value is assumed as the worst-case commutation angle error at that speed. The minimum speed of the sensorless operation depends on measurement error and the influences of the motor parameter variations. With the test-bed conditions in Fig. 8, sensorless operation down to 15 RPM is realized. If more accurate measurement devices and an A/D converter that has higher data bits are used, the minimum speed can be further decreased.

With the proposed method, the minimum speed of the sensorless operation has been improved to be around 1.5% of the rated speed.

VII. CONCLUSION

This paper presented a novel sensorless drive method for BLDC motors. This technique makes it possible to detect the rotor position over a wide speed range from near zero to high speeds. The capability of position detection at around 1.5% of the rated speed makes the starting procedure much simpler than conventional methods. Also, the proposed approach provides a precise commutation pulse even in transient state because of the speed-independent characteristic of the $G(\theta)$ function. From the simulation and experimental results, the validity of the developed sensorless drive technique using the new speed-independent function is successfully verified. Based on the successful experimental results, the proposed sensorless algorithm with wider speed range can be implemented in various industrial and other applications.

REFERENCES

[1] K. Iizuka *et al.*, "Microcomputer control for sensorless brushless motor," *IEEE Trans. Ind. Applicat.*, vol. 27, pp. 595–601, May/June 1985.
[2] J. Moreira, "Indirect sensing for rotor flux position of permanent magnet ac motors operating in a wide speed range," *IEEE Trans. Ind. Applicat.*, vol. 32, pp. 401–407, Nov./Dec. 1996.
[3] R. C. Becerra, T. M. Jahns, and M. Ehsani, "Four-quadrant sensorless brushless ECM drive," in *Proc. IEEE APEC'91 Conf.*, 1991, pp. 202–209.
[4] T. M. Jahns, R. C. Becerra, and M. Ehsani, "Integrated current regulation for a brushless ECM drive," *IEEE Trans. Power Electron.*, vol. 6, pp. 118–126, Jan. 1991.
[5] N. Ertugrul and P. Acarnley, "A new algorithm for sensorless operation of permanent magnet motors," *IEEE Trans. Ind. Applicat.*, vol. 30, pp. 126–133, Jan./Feb. 1994.
[6] R. Wu and G. R. Slemon, "A permanent magnet motor drive without a shaft sensor," *IEEE Trans. Ind. Applicat.*, vol. 27, pp. 1005–1011, Sept./Oct. 1991.
[7] S. Ogasawara and H. Akagi, "An approach to position sensorless drive for brushless dc motors," *IEEE Trans. Ind. Applicat.*, vol. 27, pp. 928–933, Sept./Oct. 1991.
[8] A. Consoli, S. Musumeci, A. Raciti, and A. Testa, "Sensorless vector and speed control of brushless motor drives," *IEEE Trans. Ind. Electron.*, vol. 41, pp. 91–96, Feb. 1994.
[9] J. P. Johnson, M. Ehsani, and Y. Guzelgunler, "Review of sensorless methods for brushless dc," in *Proc. IEEE IAS'99 Conf.*, vol. 1, 1999, pp. 143–150.
[10] M. Tomita, M. Satoh, H. Yamaguchi, S. Doki, and S. Okuma, "Sensorless estimation of rotor position of cylindrical brushless dc motors using eddy current," in *Proc. IEEE International Workshop on Advanced Motion Control*, vol. 1, 1996, pp. 24–28.
[11] T. Kim and M. Ehsani, "Advanced sensorless drive techniques for brushless dc motors," U.S. Patent 60/438,949, 2004.



Tae-Hyung Kim (S'00–M'04) received the B.S. and M.S. degrees from Korea University, Seoul, in 1994 and 1998, respectively, and the Ph.D. degree from Texas A&M University, College Station, in 2003, all in electrical engineering.

Since 2003, he has been a Postdoctoral Research Associate with the Advanced Vehicle and Motor Drives Research Center, Texas A&M University. His research interests include sensorless control of BLDC motor drives, power conversion system of BLDC generator, hybrid electric vehicles and systems, inverter topology for ac motor drives, and application of DSP controllers for motor drives.

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.



Mehrdad Ehsani (S'70–M'81–SM'83–F'96) received the Ph.D. degree in electrical engineering from the University of Wisconsin-Madison in 1981.

Since 1981, he has been with Texas A&M University, College Station, where he is now a Professor of electrical engineering and Director of the Advanced Vehicle Systems Research Program. He is the author of over 300 publications in pulsed-power supplies, high-voltage engineering, power electronics, and motor drives. He is coauthor of three books on power electronics and motor drives and a contributor to an IEEE Guide for Self-Commutated Converters and other monographs and books. He is the author of over 20 U.S. and EC patents. His current research work is in power electronics, motor drives, hybrid electric vehicles, and systems.

Dr. Ehsani received the Prize Paper Award in Static Power Converters and motor drives at the IEEE-Industry Applications Society at the 1985, 1987, and 1992 Annual Meetings, the James R. Evans Avant Garde Award from the IEEE Vehicular Technology Society, in 2001, and the IEEE Undergraduate Teaching Award in 2003. In 1992, he was named the Halliburton Professor in the College of Engineering at A&M. In 1994, he was also named the Dresser Industries Professor in the same college. In 2001 he was named the Dow Chemical Faculty Fellow of the College of Engineering at Texas A&M University. He is an Associate Editor of the IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS and the IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY. He has been a member of IEEE Power Electronics Society AdCom, past Chairman of PELS Educational Affairs Committee, past Chairman of IEEE-IAS Industrial Power Converter Committee and past chairman of the IEEE Myron Zucker Student-Faculty Grant Program. He was the General Chair of IEEE Power Electronics Specialist Conference for 1990. He has been an IEEE Industrial Electronics Society and Vehicular Technology Society Distinguished Speaker and IEEE Industry Applications Society past Distinguished Lecturer. He is Chairman of Vehicular Technology Society Vehicle Power and Propulsion Committee. He was elected to the Board of Governors of IEEE-VTS in 2003. He is also a Registered Professional Engineer in the State of Texas.

Authorized licensed use limited to: Qingdao University. Downloaded on September 16,2026 at 08:14:14 UTC from IEEE Xplore. Restrictions apply.